



Tangent and Normal : RACE-01

Single Correct Question

1. The side of a square sheet is increasing at the rate of 4 cm per minute. The rate by which the area increasing when the side is 8 cm long is-
 (A) $16 \text{ cm}^2/\text{min.}$ (B) $32 \text{ cm}^2/\text{min.}$ (C) $64 \text{ cm}^2/\text{min.}$ (D) $80 \text{ cm}^2/\text{min.}$
MTN121
2. The radius of an air bubble is increasing at the rate of 0.5 cm/sec. The rate by which the volume of the bubble is increasing when the radius is 1 cm, is-
 (A) $2\pi \text{ cm}^3/\text{s}$ (B) $\pi \text{ cm}^3/\text{s}$ (C) $4\pi \text{ cm}^3/\text{s}$ (D) $0.5\pi \text{ cm}^3/\text{s}$
MTN122
3. The point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate is-
 (A) (4, 2) (B) (-4, -2) (C) (2, 4) (D) (-2, -4)
MTN123
4. A particle moves along the curve $y = x^2 + 2x$. Then the points on the curve where the x and y coordinates of the particle changing at the same rate, are
 (A) $(-\frac{3}{4}, -\frac{1}{2})$ (B) $(-\frac{1}{2}, -\frac{3}{4})$ (C) $(\frac{3}{4}, \frac{1}{2})$ (D) $(\frac{1}{2}, \frac{3}{4})$
MTN124
5. Using differentials, find the approximate value of $\sqrt{25.2}$.
 (A) 5.02 (B) 5.01 (C) 5.03 (D) 5.04
MTN125
6. The tangent to the curve $y = ax^2 + bx$ at (2, -8) is parallel to x -axis. Then
 (A) $a = 2, b = -2$ (B) $a = 2, b = -4$ (C) $a = 2, b = -8$ (D) $a = 4, b = -4$
MTN126
7. The point of the curve $y^2 = 2(x - 3)$ at which the normal is parallel to the line $y - 2x + 1 = 0$ is
 (A) (5, 2) (B) $(-\frac{1}{2}, -2)$ (C) (5, -2) (D) $(\frac{3}{2}, 2)$
MTN127
8. At which point the line $\frac{x}{a} + \frac{y}{b} = 1$, touches the curve $y = be^{-x/a}$
 (A) (0, 0) (B) (0, a) (C) (0, b) (D) (b, 0)
MTN128
9. The area of the triangle formed by the coordinate axes and a tangent to the curve $xy = a^2$ at the point (x_1, y_1) on it is
 (A) $\frac{a^2 x_1}{y_1}$ (B) $\frac{a^2 y_1}{x_1}$ (C) $2a^2$ (D) $4a^2$
MTN129
10. An equation of the tangent to the curve $y = x^4$ from the point (2, 0) not on the curve is
 (A) $y = 0$ (B) $x = 0$ (C) $x + y = 0$ (D) None of these
MTN130

11. The sum of intercepts on co-ordinate axes made by tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ is
 (A) a (B) $2a$ (C) $2\sqrt{a}$ (D) None of these

MTN131

Multiple Correct Question

12. If tangent to curve $2y^3 = ax^2 + x^3$ at point (a, a) cuts off intercepts α, β on co-ordinate axes, where $\alpha^2 + \beta^2 = 61$, then the value of ' a ' is equal to
 (A) 20 (B) 25 (C) 30 (D) -30

MTN132

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	C	A	C	B	A	C	C	C	C	A	A	C,D



Tangent and Normal : RACE-02

Single Correct Question

1. The slope of tangent to the curve $x = t^2 + 3t - 8, y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is
 (A) $\frac{22}{7}$ (B) $\frac{6}{7}$ (C) -6 (D) None of these
 MTN133
2. If $x = t^2$ and $y = 2t$, then equation of the normal at $t = 1$ is
 (A) $x + y - 3 = 0$ (B) $x + y - 1 = 0$ (C) $x + y + 1 = 0$ (D) $x + y + 3 = 0$
 MTN134
3. The normal of the curve $x = a(\cos \theta + \theta \sin \theta), y = a(\sin \theta - \theta \cos \theta), (a \neq 0)$ at any θ is such that
 (A) It makes a constant angle with x -axis
 (B) It passes through the origin
 (C) It is at a constant distance from the origin
 (D) None of these
 MTN135
4. The normal to the curve $x = a(1 + \cos \theta), y = a \sin \theta$ at ' θ ' always passes through the fixed point
 (A) (a, a) (B) $(0, a)$ (C) $(0, 0)$ (D) $(a, 0)$
 MTN136
5. The equation of the tangent to the curve $x = 2 \cos^3 \theta$ and $y = 3 \sin^3 \theta$ at the point $\theta = \pi/4$ is
 (A) $2x + 3y = 3\sqrt{2}$ (B) $2x - 3y = 3\sqrt{2}$ (C) $3x + 2y = 3\sqrt{2}$ (D) $3x - 2y = 3\sqrt{2}$
 MTN137
6. If the tangent to the curve $x = a(\theta + \sin \theta), y = a(1 + \cos \theta)$ at $\theta = \frac{\pi}{3}$ makes an angle α ($0 \leq \alpha < \pi$) with x -axis, then $\alpha =$
 (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$
 MTN138
7. If the normal at the point $(3t, 4/t)$ of the curve $xy = 12$ cuts the curve again at $(3t_1, 4/t_1)$ then find ' t_1 ' in terms of ' t '
 (A) $\frac{-9}{16t^3}$ (B) $\frac{-16}{9t^3}$ (C) $\frac{9}{16t^3}$ (D) $\frac{16}{9t^3}$
 MTN139
8. The angle of intersection of the curves $y = x^2$ and $x = y^2$ at $(1, 1)$ is
 (A) $\tan^{-1} \left(\frac{4}{3} \right)$ (B) $\tan^{-1}(1)$ (C) 90° (D) $\tan^{-1} \left(\frac{3}{4} \right)$
 MTN140
9. If the curve $y = a^x$ and $y = b^x$ intersect at angle α then, $\tan \alpha =$
 (A) $\frac{a-b}{1+ab}$ (B) $\frac{\log a - \log b}{1 + \log a \log b}$ (C) $\frac{a+b}{1-ab}$ (D) $\frac{\log a + \log b}{1 - \log a \log b}$
 MTN141

10. For the curve $xy = c^2$ the subnormal at any point varies as
 (A) x^2 (B) x^3 (C) y^2 (D) y^3 MTN142
11. For the curve $by^2 = (x + a)^3$ the square of subtangent is proportional to
 (A) (Subnormal) $^{1/2}$ (B) Subnormal
 (C) (Subnormal) $^{3/2}$ (D) None of these MTN143
12. The subtangent, ordinate and subnormal to the parabola $y^2 = 4ax$ at a point (different from the origin) are in
 (A) AP (B) GP (C) HP (D) AGP MTN144

Multiple Correct Question

13. If at the point $\theta = \frac{\pi}{2}$ on the curve $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$, $a \neq 1$, then
 (A) length of normal = $\sqrt{2}a$ (B) Length of sub-tangent = a
 (C) length of sub-normal = a (D) Length of tangent = $\sqrt{2}a$ MTN145

Matrix Match Type Question

14. **Column - I** **Column - II**
- | | | |
|--|-----|-----|
| (A) If curves $y^2 = 4ax$ and $y = e^{-\frac{x}{2a}}$ are orthogonal then 'a' can take value | (p) | 3 |
| (B) If θ is angle between the curves $y = [\sin x + \cos x]$, ([$\cdot$] denote GIF) and $x^2 + y^2 = 5$ then $\operatorname{cosec}^2 \theta$ is | (q) | 1 |
| (C) If curves $y^2 = 4a(x + a)$ and $y^2 = 4b(x + b)$ intersects each other orthogonally then $\left \frac{a}{b} \right $ can be equal to ____ | (r) | 5/4 |
| (D) If $y = x^2 + 3x + c$ and $x = y^2 + 3y + c$ touches each other at (h, k) then $ h + k + c $ can be equal to... | (s) | 2 |
- MTN146

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	C	D	C	D	B	D	B	D
Question	11	12	13	14						
Answer	B	B	A,B,C,D	A $\rightarrow$ p,q,r,s; B $\rightarrow$ r; C $\rightarrow$ p,q,r,s; D $\rightarrow$ q						